

Course Syllabus

Fundamentals of Computer Programming (C++)

Course Code: 10740003-0

1. Course Information

Class Time and Location: Week 1 - Week 16, Wednesday 13:30, 3 sessions, Classroom 5103, Building 5

2. Course Objectives

- Learn theory and syntax through applied design requirements
- Fundamentals of modern C++ syntax, emphasizing logic and style
- Test-Driven Design

3. Course Plan

Framework: Visual Studio 2026 + Raylib Graphics Library

Key topics:

1. Hello World!
2. Data Types, Operations, and Basic Concepts of Functions
3. Basic Concepts of Classes and Objects
4. Flow Control
5. String Class, File Reading and Writing
6. Usage of Raylib Graphics Library and **Multi-File Project Structure**
7. Arrays and Vectors

8. Memory and Pointers
9. **Class Encapsulation Design** and Operator Overloading
10. **Class Inheritance**
11. Game Interface Development: Model-View-Controller (MVC) Pattern
12. Defensive Programming and Exception Handling

All the topics and schedules might be adjusted based on the actual teaching progress.

Reference book:

C++ how to program, 10th edition, Paul Deitel, Harvey Deitel

<https://item.jd.com/15099428.html>

Reference book is not mandatory but it is recommended to read it. We may not cover all the topics in the book, and some topics in the course may be more advanced than the book. But it is recommended to read the book to understand the basic concepts of C++.

4. Course Platforms

1. RainClassroom: Live classes, in-class questions, and after-class assignments

<https://pro.yuketang.cn/>

2. Feishu Documents: Course handouts, code, etc.

https://lbwcs29n19.feishu.cn/drive/folder/CVspfDf5JlDi1OdDxETcWml6nee?from=from_copylink

3. OJ Platform: Programming assignments

Will be announced later.

4. WeChat Group: Notifications and online communication

5. Web Learning: Material release and course notifications

<https://learn.tsinghua.edu.cn/>

5. Course Assessment (Grading System)

- RainClassroom Performance: 5%
- OJ Programming Assignments: 20%
- Midterm Exam (Written Test): 20%
- Final Exam (Written Test) or Final Project: 40%
- Tutorial: 15%

Note on Assessment: If the midterm exam score is not satisfactory, the final exam score can account for 60%. The final score will be the highest of the following three options:

- Midterm (20%) + Final Exam (40%)
- Midterm (20%) + Final Project (40%)
- Final Exam (60%)